

NETWORK SNIFFING

Ethical Hacking and Penetration

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INTRODUCTION

Sniffing is the process of capturing and analyzing network traffic to understand how devices communicate and to detect security issues.



ESSENTIALS

NETWORK INTERFACE CARD (NIC)

The **Network Interface Card (NIC)** is a hardware component that enables a computer to communicate over a network. It receives incoming data frames from the network medium and transmits outgoing data to other devices.

Functions of a NIC

- Connects a host to a wired or wireless network.
- Receives and transmits network frames.
- Examines the destination MAC address of each incoming frame.
- Passes valid frames to the operating system for further processing.

MEDIA ACCESS CONTROL (MAC) ADDRESS

A **Media Access Control (MAC) address** is a unique hardware identifier assigned to every Network Interface Card (NIC). It is used for communication at the Data Link Layer (Layer 2) of the OSI model.

Characteristics

- Length: 48 bits (6 bytes)
- Representation: 12 hexadecimal digits
- Example: 00:1A:2B:3C:4D:5E

Purpose

- Provides a unique identifier for every network interface.
- Enables devices within the same local network to communicate correctly.
- Allows the NIC to determine whether an incoming frame should be processed.

COLLISION DOMAIN

Collision domain is the logical and physical segment of the network where data packets can collide with one another.

When multiple hosts are interconnected via a common coaxial cable or a repeater hub, they share a singular electrical bus.

Should two nodes initiate simultaneous transmission, the resultant superposition of electrical signals—a phenomenon known as a collision—renders both frames unintelligible.

PROTOCOLS SUSCEPTIBLE TO SNIFFING

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Network protocols that transmit data without encryption (clear text) are highly vulnerable to packet sniffing. Attackers who capture network traffic can easily read sensitive information, such as usernames, passwords, emails, and application data.

APPLICATION LAYER PROTOCOLS

Several Application Layer protocols transmit sensitive information in **clear text**, making them common targets for packet sniffing.

Protocol	Information Exposed
HTTP	Web pages, form data, cookies
FTP	Username, passwords, transferred files
TFTP	Files transferred without encryption
Telnet	Username, passwords, commands
SMTP	Email messages
POP3	Email username, passwords, messages

Note: Modern secure alternatives (e.g., HTTPS, SFTP, SSH) encrypt transmitted data and significantly reduce the risk of sniffing.

TRANSPORT LAYER INFORMATION

The **Transport Layer** provides valuable information that helps analyze network communications and identify active services.

TCP	UDP
• Source and destination port numbers. • Sequence and acknowledgment numbers. • Connection establishment and termination. • Session tracking and analysis.	• Source and destination port numbers. • Lightweight communication without connection establishment. • Commonly used for DNS, VoIP, and streaming services.

ADDRESS RESOLUTION PROTOCOL (ARP)

ADDRESS RESOLUTION PROTOCOL (ARP)

The Address Resolution Protocol (ARP) is a Data Link Layer protocol used to map an IPv4 address to its corresponding Media Access Control (MAC) address within a local network.

HOW ARP WORKS

ARP resolves the destination MAC address through two message types:

ARP Request	ARP Reply
• Broadcast message sent to all devices on the local network. • Asks: "Who has this IP address?"	• Unicast response sent by the device that owns the requested IP address. • Returns its corresponding MAC address.

Communication Process

- The sender determines the destination IP address.
- An ARP Request is broadcast on the local network.
- The destination device sends an ARP Reply containing its MAC address.
- The sender builds the Ethernet frame and transmits the data.

ACTIVE AND PASSIVE SNIFFING

TYPES OF PACKET SNIFFING

Packet sniffing can be classified into two main categories based on the network environment and the method used to capture traffic.

Passive Sniffing	Active Sniffing
• Captures network traffic without modifying network communications. • Commonly performed in hub-based networks. • Does not generate additional network traffic.	• Captures traffic in switched networks. • Requires techniques to redirect or duplicate packets. • Involves interaction with network devices or traffic manipulation.

PASSIVE SNIFFING

Passive sniffing is the process of monitoring and capturing network traffic without transmitting or modifying packets.

Characteristics

- Operates silently without affecting network communication.
- Requires the attacker to be within the same collision domain as the target.
- Most effective in hub-based Ethernet networks, where all connected devices receive the same traffic.

Advantages	Limitations
• Difficult to detect. • Simple to perform. • Does not introduce additional network traffic	• Ineffective in modern switched Ethernet networks. • Limited to traffic visible within the local collision domain.

ACTIVE SNIFFING

Active sniffing is the process of capturing traffic in switched networks by influencing or manipulating network communication.

Characteristics

- Used when passive sniffing is not possible.
- Requires interaction with network devices or packet manipulation.
- Commonly performed in switch-based Ethernet networks.

Advantages	Limitations
• Enables monitoring of traffic in switched networks. • Suitable for network analysis and security testing.	• More complex than passive sniffing. • May generate detectable network activity. • Often requires administrative access or successful attack techniques.

ACTIVE SNIFFING

Common Techniques for Active Sniffing

- Port Mirroring (SPAN Port): Configures a switch to copy traffic from one or more ports to a monitoring port.
- ARP Spoofing (ARP Poisoning): Redirects traffic through the attacker's device.
- MAC Flooding: Overloads the switch's MAC address table to force broadcast behavior.

SNIFFING TOOLS AND TECHNIQUES

PACKET SNIFFING TOOLS

Packet sniffing tools capture and analyze network traffic to support network monitoring, troubleshooting, and security assessment.

Tool	Primary Purpose
Wireshark	Packet capture and protocol analysis
Ettercap	Network monitoring and man-in-the-middle attacks
EtherPeek	Network traffic analysis
Snort	Intrusion Detection System (IDS) with packet capture capabilities

ACTIVE SNIFFING TECHNIQUES

In modern switched networks, passive sniffing is usually ineffective because switches forward frames only to the intended destination. Active sniffing techniques are used to gain access to network traffic.

Common Techniques

MAC
Flooding

ARP Spoofing
(ARP Poisoning)

Port Mirroring
(SPAN Port)

DHCP
Starvation

MAC FLOODING

MAC Flooding is an active sniffing technique that overwhelms a switch by sending a large number of frames with spoofed MAC addresses.

How it works

- The attacker transmits thousands of frames with fake MAC addresses.
- The switch stores each new address in its CAM table.
- The CAM table becomes full.
- The switch can no longer learn new MAC addresses.
- Unknown frames are flooded to all switch ports.

CONTENT ADDRESSABLE MEMORY (CAM) TABLE

A Content-Addressable Memory (CAM) table is a database maintained by an Ethernet switch to associate MAC addresses with switch ports. This enables the switch to forward frames only to the intended destination.

Relationship to MAC Flooding

MAC flooding targets the CAM table by inserting a large number of fake MAC addresses until the table reaches its capacity.

- The switch cannot learn new MAC addresses.
- Unknown frames cannot be forwarded normally.
- The switch floods these frames to all ports, behaving similarly to a hub.
- An attacker can capture traffic intended for other devices.

ARP SPOOFING (ARP POISONING)

ARP Spoofing is an active sniffing technique in which an attacker sends forged ARP messages to associate their MAC address with the IP address of another device, such as the default gateway.

The Process:



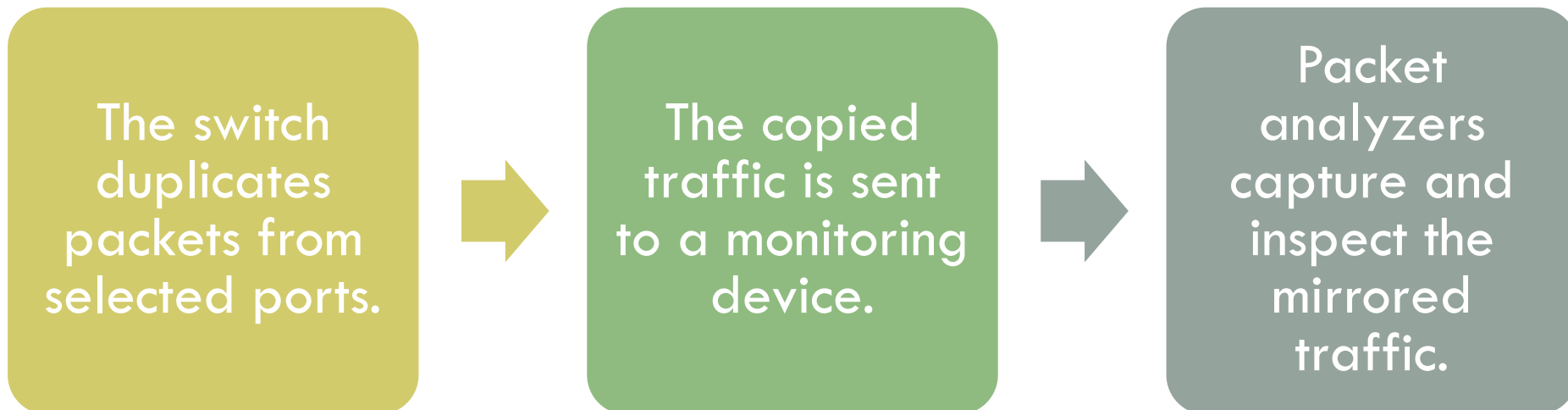
THE IMPACT OF ARP SPOOFING

- Enables **Man-in-the-Middle (MitM)** attacks.
- Allows packet sniffing in switched networks.
- May lead to data theft or session hijacking.

PORT MIRRORING (SPAN PORT)

Port Mirroring, also known as a Switched Port Analyzer (SPAN), is a switch feature that copies network traffic from one or more ports to a designated monitoring port.

The process



DHCP STARVATION ATTACK

A DHCP Starvation Attack is a Denial-of-Service (DoS) attack in which an attacker exhausts all available IP addresses in a DHCP server's address pool.

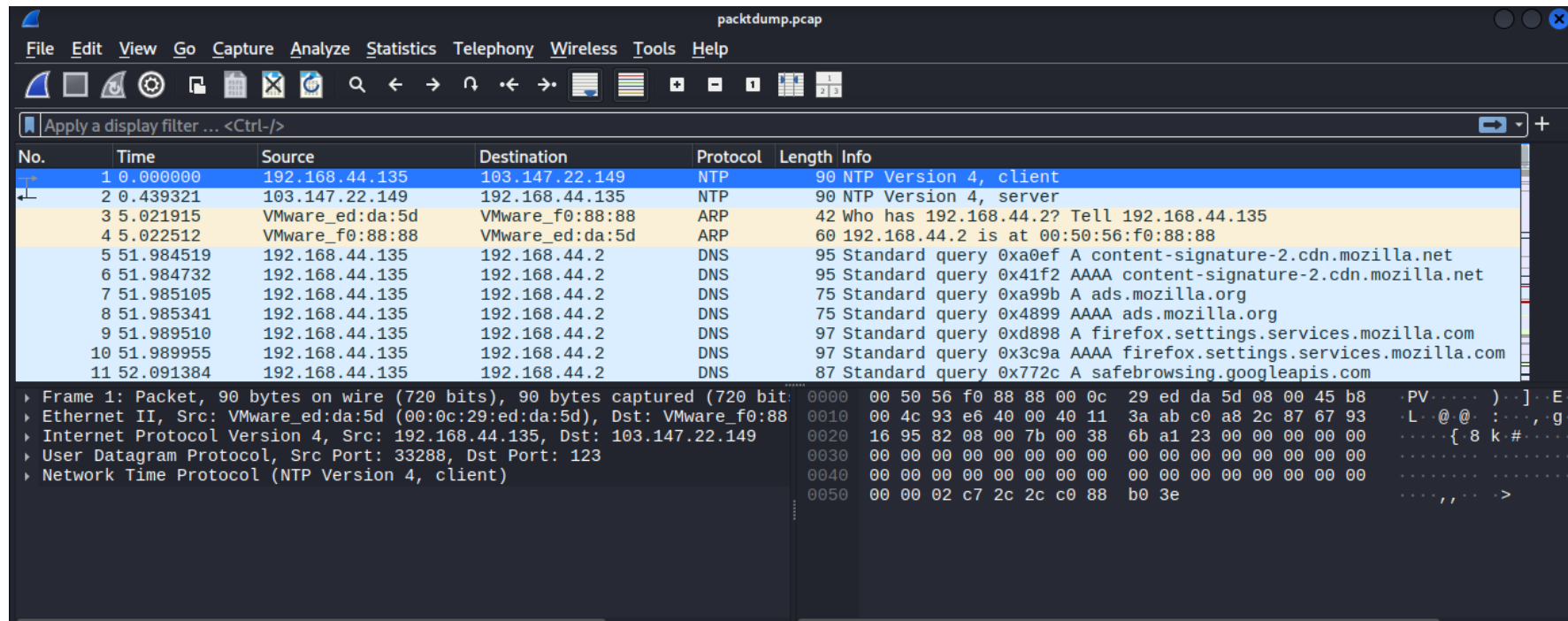
Attack Process

- The attacker sends numerous fake DHCP Discover/Request messages.
- The DHCP server assigns IP addresses to nonexistent clients.
- The address pool becomes exhausted.
- Legitimate users cannot obtain or renew IP addresses.

SNIFFING TOOLS

WIRESHARK

Wireshark is an open-source network protocol analyzer used to capture, inspect, and analyze network traffic in real time. It is one of the most widely used packet sniffing tools for network administration, troubleshooting, and security analysis.



WIRESHARK FILTERS

Filter	Description
arp	Display ARP packets
icmp	Display ICMP packets
dns	Display DNS packets
http.request	Display HTTP requests
tcp.port == 23	Display Telnet traffic
ip.addr == 192.168.1.10	Display packets associated with a specific IP
tcp contains "password"	Display TCP packets containing a specific string

Network Packet Capture Features



TCPDUMP

Tcpdump is a command-line packet analyzer commonly used on Linux and Unix systems to capture and inspect network traffic

Command	Function
<code>tcpdump -i eth0</code>	Capture packets on interface eth0
<code>tcpdump -w capture.pcap</code>	Save captured packets to a file
<code>tcpdump -r capture.pcap</code>	Read packets from a capture file

```
(kali㉿kali)-[~]  
└─$ sudo tcpdump -i eth0 -s 0 -w packtdump.pcap  
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```


TCPDUMP

Features

- Captures packets from a specified network interface.
- Supports packet filtering using Boolean expressions.
- Displays packet information in real time.
- Saves captured packets for later analysis.

